A product counterexample to the simplex maximum for projection-body volume

Rewritten for readability from the OpenAI manuscript of the same title

Abstract

The product of two ten-dimensional simplices has larger normalized projection-body volume than a twenty-dimensional simplex. This gives a counterexample to Brannen's proposed simplex maximum. The proof is elementary: a facet formula for projection bodies of polytopes, a product identity, the exact simplex value, and one integer inequality.

1 The result

1.1 Definitions

Throughout, $d \geq 2$. A *convex body* in $\mathbb{R}^d$ is a compact convex set with nonempty interior. We write $|K|$ for the d-dimensional volume of $K \subset \mathbb{R}^d$, and vol_j for j-dimensional volume inside a j-dimensional affine subspace. For a compact convex set C, its *support function* is

$$h_C(x) = \max_{y \in C} \langle x, y \rangle, \qquad x \in \mathbb{R}^d.$$

S^{d-1} is the Euclidean unit sphere, $\mathrm{proj}_{u^\perp}$ is the orthogonal projection onto the hyperplane $u^\perp$, B_2^j is the Euclidean unit ball of $\mathbb{R}^j$, and $\kappa_j = |B_2^j|$.

The *projection body* ΠK (also $\Pi_d K$) of a convex body $K \subset \mathbb{R}^d$ is the compact convex set whose support function is

$$h_{\Pi K}(u) = \mathrm{vol}_{d-1}\big(\mathrm{proj}_{u^\perp} K\big) \quad (u \in S^{d-1}), \qquad h_{\Pi K}(tu) = t\, h_{\Pi K}(u) \quad (t \geq 0).$$

A compact convex set is determined by its support function (fact (S2) in Section 2). That a set with this support function exists is proved in Lemma 2 for polytopes, which is all the main theorem needs, and in Part B of the proof of Lemma 4 for general convex bodies.

The functional under study, and the constant it is compared with, are

$$R_d(K) = \frac{|\Pi K|}{|K|^{d-1}}, \qquad c_d = \frac{(d+1)d^d}{d!}. \tag{1}$$

Proposition 5 shows that $R_d(S) = c_d$ for every full-dimensional simplex $S \subset \mathbb{R}^d$. The statement "simplices maximize R_d" is therefore the inequality

$$|\Pi K| \leq c_d |K|^{d-1} \qquad \text{for all convex bodies } K \subset \mathbb{R}^d. \tag{2}$$

1.2 Main theorem

Theorem 1. *Let $T_{10} = \operatorname{conv}(0, e_1, \dots, e_{10}) \subset \mathbb{R}^{10}$, where the e_i are the coordinate unit vectors, and let $K = T_{10} \times T_{10} \subset \mathbb{R}^{20}$. Then*

$$\frac{R_{20}(K)}{c_{20}} = \frac{121\binom{20}{10}}{21 \cdot 2^{20}} = \frac{22\,355\,476}{22\,020\,096} > 1.$$

Thus K violates (2) in dimension twenty.

A quantitative extension, Corollary 6, shows that products of simplices beat the simplex value by a factor λ^n, with a fixed $\lambda > 1$, in every sufficiently large dimension n.

1.3 Context

The statements in this subsection are background quoted from the source with its citations. None of them is used in a proof below.

Petty established the affine covariance of the projection-body operator [6]; see also [5, Equation (1)]. Brannen conjectured in 1996 that simplices maximize R_d among all convex bodies [1], which is (2). It is the volume of ΠK itself that occurs here, not the volume of its polar: the classical projection inequalities involving the polar concern a different functional. For $d \geq 3$, Petty's separate minimum problem asks whether ellipsoids are the only minimizers of the same functional R_d; see [7, Section 1] for these distinctions.

In dimension three, Chen, Feng, Li, Xi, and Xu settle the minimum problem and also prove $R_3(K) \leq 18 = c_3$ for all three-dimensional convex bodies, with tetrahedra attaining equality [2, Theorems 1.1 and 1.2]. Feng, Hu, Liu, and Xu give counterexamples to (2) in every dimension $d \geq 9$, using polytopes with at most $d+2$ facets [3, Theorem 1.3]; their construction uses perturbed projections of a cube and Minkowski's existence theorem [3, Section 4]. So the unrestricted-dimensional simplex-maximum question already has a negative answer. The present paper supplies a direct Cartesian-product construction with an exact value. For centrally symmetric bodies, Henk gives the lower bound $2^d(9/8)^{\lfloor d/3 \rfloor}$ for the optimal upper constant [4]. The optimal upper constant in (2) and its maximizing bodies remain separate questions.

1.4 Proof at a glance

Key idea. The functional R_d is multiplicative under Cartesian products of polytopes: $R_{r+s}(A \times B) = R_r(A)R_s(B)$. The simplex constants are not: $c_{10}^2 > c_{20}$. Hence $T_{10} \times T_{10}$ beats the 20-simplex.

Steps.

1. *Facet formula* (Lemma 2). For a polytope P, ΠP is a Minkowski sum of segments, one segment $[-\frac{1}{2}s_F\nu_F, \frac{1}{2}s_F\nu_F]$ per facet F (area s_F, outward unit normal ν_F).

2. *Product identity* (Proposition 3). The facets of $A \times B$ are $F \times B$ and $A \times G$. With step 1 this gives $\Pi(A \times B) = (|B|\Pi A) \times (|A|\Pi B)$, and multiplicativity of R follows by counting exponents.

3. *Affine invariance* (Lemma 4). $R_d(LP) = R_d(P)$ for invertible linear L; translations also preserve R_d. For polytopes this follows from step 1.

4. *Simplex value* (Proposition 5). By step 1, ΠT_d is a translate of $([0,1]^d + [0,w])/(d-1)!$ with $w = (1, \dots, 1)$. A Fubini argument gives $|[0,1]^d + [0,w]| = d+1$, so $R_d(T_d) = c_d$. Step 3 extends this to all simplices.

5. *Arithmetic* (Section 7). Steps 2 and 4 give $R_{20}(K) = c_{10}^2$, and c_{10}^2/c_{20} is the stated fraction.

Dependencies. Lemma 2 feeds Proposition 3, Part A of Lemma 4, and Proposition 5. Lemma 4 is used only for the last sentence of Proposition 5 (all simplices have the same value); its Part B (non-polytopal bodies) is used nowhere else. Theorem 1 and Corollary 6 use only Propositions 3 and 5 plus explicit arithmetic.

2 Conventions and tools

Renamed symbols. To keep one meaning per symbol, four letters of the source are changed. In the proof of Lemma 4, the radius R becomes ρ, the functions F, f_m become g, g_m, and the factors λ_m become τ_m. In Corollary 6, the constant A becomes Θ, and the residue r with $d_r = 10 + r$ becomes i with $q = 10 + i$.

Minkowski sum. $C + D = \{c + d : c \in C,\ d \in D\}$. For $v \in \mathbb{R}^d$, $[a, b]$ denotes the segment with endpoints a, b.

Support-function facts. Let C, D be compact convex sets.

(S1) $h_{C+D} = h_C + h_D$; $h_{tC} = t\,h_C$ for $t \geq 0$; $h_{MC}(u) = h_C(M^t u)$ for a linear map M; and for $C \subset \mathbb{R}^r$, $D \subset \mathbb{R}^s$, $h_{C\times D}(u, v) = h_C(u) + h_D(v)$. Each is immediate from the definition: for instance $\max_{c,d}\langle u, c + d\rangle = \max_c\langle u, c\rangle + \max_d\langle u, d\rangle$ and $\langle u, My\rangle = \langle M^t u, y\rangle$.

(S2) If $h_C \leq h_D$ everywhere, then $C \subset D$. In particular $h_C = h_D$ implies $C = D$.

Proof. Suppose $x \notin D$ and let $y \in D$ be a point of D nearest to x. For $z \in D$ and $t \in (0, 1]$ the point $y+t(z-y)$ lies in D, so $\|x-y\|^2 \leq \|x-y-t(z-y)\|^2 = \|x-y\|^2-2t\langle x-y, z-y\rangle+t^2\|z-y\|^2$. Dividing by t and letting $t \to 0$ gives $\langle x - y, z - y\rangle \leq 0$. Hence $h_D(x - y) = \langle x - y, y\rangle < \langle x-y, y\rangle+\|x-y\|^2 = \langle x-y, x\rangle$. If x were in C we would get $h_C(x-y) \geq \langle x-y, x\rangle > h_D(x-y)$, contrary to hypothesis. So $C \subset D$. □

(S3) $h_{[-v/2,\,v/2]}(u) = \frac{1}{2}|\langle v, u\rangle|$, because a linear function on a segment attains its maximum at an endpoint.

Polytope facts. A *convex polytope* is the convex hull of finitely many points. For a full-dimensional polytope $P \subset \mathbb{R}^d$ and $u \in \mathbb{R}^d$, the *face exposed by* u is $P(u) = \{x \in P : \langle u, x\rangle = h_P(u)\}$; thus $P(0) = P$, and for $u \neq 0$ the face $P(u)$ lies in a hyperplane, so $\dim P(u) \leq d - 1$. A *facet* is a face of dimension $d - 1$; it equals $P(\nu)$ for its outward unit normal ν. We use the standard facts that P has finitely many faces, that the boundary of P is the union of its facets, and that two distinct facets meet in a face of dimension at most $d - 2$.

Volume facts. $|MC| = |\det M|\,|C|$ for linear M; $|tC| = t^d|C|$ in $\mathbb{R}^d$; $|C \times D| = |C|\,|D|$.

3 The facet formula

This lemma is the polytopal case of Cauchy's projection formula; see [7, Section 1]. It is proved here in full because the factor $\frac{1}{2}$ fixes all later constants. It is used in Proposition 3, Lemma 4 and Proposition 5.

Lemma 2. *Let $P \subset \mathbb{R}^d$, $d \geq 2$, be a full-dimensional convex polytope. For each facet F, let s_F be its $(d-1)$-dimensional volume and ν_F its outward unit normal. Then*

$$h_{\Pi P}(u) = \frac{1}{2}\sum_F s_F|\langle \nu_F, u\rangle|, \qquad u \in \mathbb{R}^d, \tag{3}$$

and therefore

$$\Pi P = \sum_F \left[-\frac{s_F \nu_F}{2}, \frac{s_F \nu_F}{2} \right]. \tag{4}$$

Proof. Plan. For a unit vector u, project each facet to $u^\perp$. The projected facet has volume $s_F|\langle \nu_F, u\rangle|$ (Step 1), and almost every point of the shadow of P is covered by exactly two projected facets (Step 2). Integrating gives (3) (Step 3). Fix a unit vector u and write $\pi = \mathrm{proj}_{u^\perp}$.

Step 1 (Jacobian). *Claim: for every facet F, $\mathrm{vol}_{d-1}(\pi F) = s_F|\langle \nu_F, u\rangle|$. If $\langle \nu_F, u\rangle = 0$, then πF lies in an affine subspace of $u^\perp$ of dimension at most $d-2$.*

Let $v_1, \dots, v_{d-1}$ be an orthonormal basis of $\nu_F^\perp$, the direction space of the hyperplane H of F. The restriction $\pi|_H \colon H \to u^\perp$ is an affine map between $(d-1)$-dimensional spaces, so it multiplies $(d-1)$-volume by the $(d-1)$-volume J of the parallelepiped spanned by $\pi v_1, \dots, \pi v_{d-1}$. Since u is a unit vector orthogonal to all πv_i, J equals the d-volume of the parallelepiped spanned by $\pi v_1, \dots, \pi v_{d-1}, u$:

$$J = |\det(\pi v_1, \dots, \pi v_{d-1}, u)| = |\det(v_1, \dots, v_{d-1}, u)|,$$

where the second equality holds because $\pi v_i = v_i - \langle v_i, u\rangle u$ differs from v_i by a multiple of the last column. Expand $u = \langle u, \nu_F\rangle \nu_F + \sum_i \langle u, v_i\rangle v_i$ in the last column. The terms with v_i repeat a column and vanish, so $J = |\langle u, \nu_F\rangle| \cdot |\det(v_1, \dots, v_{d-1}, \nu_F)| = |\langle \nu_F, u\rangle|$, the last matrix being orthogonal. If $J \neq 0$, $\pi|_H$ is an affine bijection and $\mathrm{vol}_{d-1}(\pi F) = J s_F$. If $J = 0$, the linear part of $\pi|_H$ is singular, so πH is a proper affine subspace of $u^\perp$ and $\mathrm{vol}_{d-1}(\pi F) = 0 = J s_F$.

Step 2 (multiplicity two). Let $E \subset u^\perp$ be the union of

(i) the relative boundary of πP in $u^\perp$;
(ii) the sets πG for all faces G of P with $\dim G \leq d-2$;
(iii) the sets πF for all facets F with $\langle \nu_F, u\rangle = 0$.

E has $(d-1)$-dimensional measure zero. The boundary of a convex set is a null set, which covers (i). The sets in (ii) and (iii) are finitely many and each lies in an affine subspace of dimension at most $d-2$ (for (iii) by Step 1).

Claim: every $y \in \pi P \setminus E$ lies in πF for exactly two facets F. Let $\ell = y + \mathbb{R}u$.

(a) *ℓ meets the interior of P.* Otherwise the separation theorem gives $\xi \neq 0$ and $c \in \mathbb{R}$ with $\langle \xi, x\rangle \leq c \leq \langle \xi, z\rangle$ for all $x \in P$, $z \in \ell$. Boundedness below on the line ℓ forces $\langle \xi, u\rangle = 0$, so $\xi \in u^\perp$ and $\langle \xi, \pi x\rangle = \langle \xi, x\rangle \leq c \leq \langle \xi, y\rangle$ for all $x \in P$. Then y maximizes a nonzero linear functional of $u^\perp$ over πP, so y is a relative boundary point of πP, contradicting (i).

(b) By (a) and convexity, $\ell \cap P$ is a segment $[p^-, p^+]$ with $p^- \neq p^+$. Its endpoints lie in the boundary of P and all its other points lie in the interior of P. (If c is an interior point and $p \in P$, every point of $[c, p]$ other than p is interior.) Thus ℓ meets the boundary of P exactly in $\{p^-, p^+\}$.

(c) The boundary point p^- lies in some facet. It lies in no face of dimension at most $d-2$, since $\pi p^- = y$ avoids the sets (ii). As two distinct facets meet in such a face, p^- lies in exactly one facet F^-. Likewise p^+ lies in exactly one facet F^+.

(d) $F^- \neq F^+$: a facet containing both p^- and p^+ has a hyperplane containing the direction $p^+ - p^-$, a nonzero multiple of u, so $\langle \nu_F, u\rangle = 0$ and $y \in \pi F \subset E$ by (iii), a contradiction.

(e) If $y \in \pi F$ for a facet F, then F contains a point of ℓ, which is a boundary point of P, hence p^- or p^+ by (b). So $F \in \{F^-, F^+\}$ by (c). Conversely $y \in \pi F^\pm$. With (d), exactly two facets qualify.

Step 3 (integration). Write 1_X for the indicator function of a set X. Each $\pi F \subset \pi P$, so by Step 1, Step 2 and the nullity of E,

$$\sum_F s_F|\langle \nu_F, u\rangle| = \sum_F \mathrm{vol}_{d-1}(\pi F) = \int_{u^\perp} \sum_F 1_{\pi F}(y)\, dy = \int_{\pi P \setminus E} 2\, dy = 2\,\mathrm{vol}_{d-1}(\pi P).$$

This is (3) for unit u. Both sides of (3) are positively homogeneous of degree one in u (the left side by definition) and vanish at $u = 0$, so (3) holds for all $u \in \mathbb{R}^d$.

Step 4 (zonotope). By (S3) and (S1), the right side of (3) is the support function of the Minkowski sum in (4). So a compact convex set with the support function required of ΠP exists, and by (S2) it is that sum. □

4 The product identity

This proposition is the mechanism of the counterexample. It is used in the proofs of Theorem 1 and Corollary 6.

Proposition 3. *Let $A \subset \mathbb{R}^r$ and $B \subset \mathbb{R}^s$ be full-dimensional convex polytopes, where $r, s \geq 2$. In the orthogonal product space $\mathbb{R}^{r+s} = \mathbb{R}^r \times \mathbb{R}^s$,*

$$\Pi(A \times B) = (|B|\Pi A) \times (|A|\Pi B). \tag{5}$$

In particular,

$$R_{r+s}(A \times B) = R_r(A) R_s(B). \tag{6}$$

Proof. Plan. Identify the facets of $A \times B$ with their areas and normals, insert them into the facet formula (3), and read off a product. Then compare volumes.

Step 1 (facets of the product). *Claim: the facets of $A \times B$ are exactly the sets $F \times B$ with F a facet of A, and $A \times G$ with G a facet of B.*

For $(u, v) \in \mathbb{R}^r \times \mathbb{R}^s$ and $(x, y) \in A \times B$ we have $\langle (u, v), (x, y) \rangle = \langle u, x \rangle + \langle v, y \rangle$, which is maximal exactly when both terms are. Hence the exposed faces satisfy $(A \times B)(u, v) = A(u) \times B(v)$, and $\dim(A \times B)(u, v) = \dim A(u) + \dim B(v)$. A facet of $A \times B$ is $(A \times B)(u, v)$ for some $(u, v) \neq 0$ and has dimension $r + s - 1$. If $u \neq 0$ and $v \neq 0$, then $\dim A(u) \leq r - 1$ and $\dim B(v) \leq s - 1$, so the dimension is at most $r + s - 2$; this case is impossible. If $v = 0 \neq u$, the face is $A(u) \times B$, of dimension $\dim A(u) + s$, so $A(u)$ is a facet F of A. The case $u = 0 \neq v$ is symmetric. Conversely $F \times B = (A \times B)(\nu_F, 0)$ has dimension $r + s - 1$, so it is a facet, and likewise $A \times G$.

Step 2 (normals and areas). The facet $F \times B$ is exposed by the unit vector $(\nu_F, 0)$, which is therefore its outward unit normal. It lies in the hyperplane $\mathrm{aff}(F) \times \mathbb{R}^s$, which is the orthogonal product of the hyperplane $\mathrm{aff}(F) \subset \mathbb{R}^r$ and $\mathbb{R}^s$, so its $(r + s - 1)$-volume is the product $s_F |B|$. Symmetrically, $A \times G$ has outward unit normal $(0, \nu_G)$ and volume $|A| s_G$. The area-normal vectors are thus

$$(|B| s_F \nu_F, 0) \quad \text{and} \quad (0, |A| s_G \nu_G).$$

Step 3 (support function). By (3) applied to $A \times B$, then to A and to B,

$$h_{\Pi(A \times B)}(u, v) = \frac{1}{2} \sum_F |B| s_F |\langle \nu_F, u \rangle| + \frac{1}{2} \sum_G |A| s_G |\langle \nu_G, v \rangle| = |B| h_{\Pi A}(u) + |A| h_{\Pi B}(v).$$

By (S1) this is the support function of $(|B|\Pi A) \times (|A|\Pi B)$, so (5) follows from (S2).

Step 4 (volumes). Since $\Pi A \subset \mathbb{R}^r$ and $\Pi B \subset \mathbb{R}^s$, (5) gives $|\Pi(A \times B)| = |B|^r |\Pi A| \cdot |A|^s |\Pi B|$. Also $|A \times B| = |A||B|$. Hence

$$R_{r+s}(A \times B) = \frac{|B|^r |A|^s |\Pi A||\Pi B|}{(|A||B|)^{r+s-1}} = \frac{|\Pi A|}{|A|^{r-1}} \cdot \frac{|\Pi B|}{|B|^{s-1}} = R_r(A) R_s(B),$$

the exponents being $s - (r + s - 1) = -(r - 1)$ for $|A|$ and $r - (r + s - 1) = -(s - 1)$ for $|B|$. □

5 Affine invariance

To speak of "the simplex value" we need all simplices of a given dimension to have the same R_d. The following lemma supplies this. It is used only in the last sentence of Proposition 5, and there only for polytopes (Part A below).

Lemma 4. *Let $P \subset \mathbb{R}^d$, $d \geq 2$, be a convex body. For every invertible linear map $L\colon \mathbb{R}^d \to \mathbb{R}^d$,*

$$\Pi(LP) = |\det L| L^{-t} \Pi P, \qquad R_d(LP) = R_d(P),$$

where L^{-t} denotes the inverse transpose. Translations also preserve R_d.

Proof. Plan. Translations are immediate. Part A proves the lemma for polytopes by transforming each facet's area-normal vector and applying (4). Part B extends it to arbitrary convex bodies by inscribed polytopes, and also proves that ΠP exists.

Translations. For $t \in \mathbb{R}^d$, $\mathrm{proj}_{u^\perp}(P + t)$ is a translate of $\mathrm{proj}_{u^\perp} P$, so the two bodies have the same projection volumes, hence the same projection body, and $|P + t| = |P|$. So $R_d(P + t) = R_d(P)$.

Part A: P is a polytope.

A1 (normals). L is a linear bijection, so the facets of LP are the sets LF, F a facet of P. For $x \in P$, $\langle L^{-t}\nu_F, Lx\rangle = \langle \nu_F, x\rangle$. Hence $L^{-t}\nu_F$ is maximized over LP exactly on LF, and the outward unit normal of LF is $\nu_{LF} = L^{-t}\nu_F / \|L^{-t}\nu_F\|$. This holds for either sign of $\det L$.

A2 (areas). *Claim:* $s_{LF} = |\det L| \, \|L^{-t}\nu_F\| \, s_F$. Let $Z = F + [0, \nu_F]$, a prism of height one over F; $|Z| = s_F$. Its image is $LZ = LF + [0, L\nu_F]$, with $|LZ| = |\det L| s_F$. The slices of LZ parallel to LF are the translates $LF + tL\nu_F$, $t \in [0, 1]$, at distance $t\langle L\nu_F, \nu_{LF}\rangle = t\langle \nu_F, \nu_F\rangle / \|L^{-t}\nu_F\| = t/\|L^{-t}\nu_F\|$ from the hyperplane of LF. By Fubini, $|LZ| = s_{LF}/\|L^{-t}\nu_F\|$. Equating the two expressions for $|LZ|$ gives the claim.

A3 (covariance). By A1 and A2, $s_{LF}\nu_{LF} = |\det L| \, L^{-t}(s_F \nu_F)$. Put $M = |\det L| L^{-t}$. A linear map sends segments to segments and Minkowski sums to Minkowski sums, so (4) gives

$$\Pi(LP) = \sum_F \left[-\tfrac{1}{2}M(s_F\nu_F), \tfrac{1}{2}M(s_F\nu_F)\right] = M\,\Pi P.$$

A4 (invariance). $|\det M| = |\det L|^d |\det L|^{-1} = |\det L|^{d-1}$, so $|\Pi(LP)| = |\det L|^{d-1}|\Pi P|$. Also $|LP|^{d-1} = |\det L|^{d-1}|P|^{d-1}$. Dividing, $R_d(LP) = R_d(P)$.

Part B: P is an arbitrary convex body. (Not needed for Theorem 1 or Corollary 6.)

B1 (approximating polytopes). By translation invariance we may assume that 0 is an interior point of P. Choose $\rho > 0$ with $P \subset \rho B_2^d$, and a polytope $P_0 \subset P$ containing a ball aB_2^d, $a > 0$ (for instance a small cube centered at 0). For each $m \geq 1$ choose a finite set $N_m \subset P$ such that every point of P is within $\varepsilon_m := 1/m$ of N_m (compactness), and let P_m be the convex hull of $P_0 \cup N_1 \cup \cdots \cup N_m$. Then the P_m are polytopes, $aB_2^d \subset P_m \subset P_{m+1} \subset P$, and

$$P_m \subset P \subset P_m + \varepsilon_m B_2^d \subset \tau_m P_m, \qquad \tau_m := 1 + \varepsilon_m/a \longrightarrow 1.$$

The last inclusion holds because $\varepsilon_m B_2^d = (\varepsilon_m/a)\, aB_2^d \subset (\varepsilon_m/a)P_m$ and $P_m + (\varepsilon_m/a)P_m = \tau_m P_m$ by convexity.

B2 (projection volumes). For unit u put $g(u) = \mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} P)$ and $g_m(u) = h_{\Pi P_m}(u) = \mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} P_m)$. Projecting $P_m \subset P \subset \tau_m P_m$ and taking $(d-1)$-volumes gives

$$g_m(u) \leq g(u) \leq \tau_m^{d-1} g_m(u). \tag{7}$$

Since $\mathrm{proj}_{u^\perp} P_m$ lies in a $(d-1)$-ball of radius ρ, $g_m(u) \le \kappa_{d-1}\rho^{d-1}$, and therefore

$$0 \le g(u) - g_m(u) \le (\tau_m^{d-1} - 1)\kappa_{d-1}\rho^{d-1}.$$

Thus $g_m \to g$ uniformly on S^{d-1}. The sequence g_m is nondecreasing, because $P_m \subset P_{m+1}$.

B3 (existence of ΠP). By (S2) and monotonicity of g_m, the bodies ΠP_m increase. They lie in the ball $\kappa_{d-1}\rho^{d-1}B_2^d$ (as $g_m \le \kappa_{d-1}\rho^{d-1}$) and contain $\kappa_{d-1}a^{d-1}B_2^d$ (as $g_m(u) \ge \mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} aB_2^d) = \kappa_{d-1}a^{d-1}$), again by (S2). The union of an increasing family of convex sets is convex, so the closure Z of $\bigcup_m \Pi P_m$ is a convex body. For unit u, $h_Z(u) = \sup_m h_{\Pi P_m}(u) = \sup_m g_m(u) = g(u)$ by B2. Hence the homogeneous extension of g is the support function of Z; that is, ΠP exists and equals Z.

B4 (volume control). By (7), (S1) and (S2),

$$\Pi P_m \subset \Pi P \subset \tau_m^{d-1}\Pi P_m, \qquad |\Pi P_m| \le |\Pi P| \le \tau_m^{d(d-1)}|\Pi P_m|,$$

and from B1, $|P_m| \le |P| \le \tau_m^d |P_m|$. Consequently $|\Pi P_m| \to |\Pi P|$, $|P_m| \to |P|$, and $R_d(P_m) \to R_d(P)$.

B5 (the same for LP). Applying L to B1 gives polytopes LP_m, increasing, with $LP_m \subset LP \subset \tau_m LP_m$, $LP \subset \|L\|\rho B_2^d$, and $LP_m \supset (a/\|L^{-1}\|)B_2^d$. Steps B2–B4 used only these properties, so they hold with LP, LP_m in place of P, P_m. In particular $h_{\Pi(LP_m)} \to h_{\Pi(LP)}$ pointwise and $R_d(LP_m) \to R_d(LP)$.

B6 (limit). By Part A, $\Pi(LP_m) = |\det L| L^{-t}\Pi P_m$, so by (S1), for every u,

$$h_{\Pi(LP_m)}(u) = |\det L|\, h_{\Pi P_m}(L^{-1}u).$$

Letting $m \to \infty$ (B2 for P, B5 for LP, and homogeneity) gives $h_{\Pi(LP)}(u) = |\det L|\, h_{\Pi P}(L^{-1}u)$, which by (S1) is the support function of $|\det L| L^{-t}\Pi P$. By (S2), $\Pi(LP) = |\det L| L^{-t}\Pi P$. Finally $R_d(LP_m) = R_d(P_m)$ by Part A, and the limits in B4 and B5 give $R_d(LP) = R_d(P)$. □

6 The simplex value

This proposition gives the number with which the product is compared. It is used in the proofs of Theorem 1 (with $d = 10$, and with $d = 20$ to interpret c_{20}) and of Corollary 6.

Proposition 5. *For $d \ge 2$, the standard simplex $T_d = \mathrm{conv}(0, e_1, \dots, e_d) \subset \mathbb{R}^d$ satisfies*

$$|T_d| = \frac{1}{d!}, \qquad |\Pi T_d| = \frac{d+1}{((d-1)!)^d}, \qquad R_d(T_d) = c_d.$$

Every full-dimensional d-simplex has the same value c_d.

Proof. Plan. List the $d+1$ area-normal vectors of T_d. By (4), ΠT_d is a scaled translate of the unit cube plus one segment, whose volume is found by slicing along that segment. Write $w = e_1 + \cdots + e_d$ and $T_d = \{x : x_i \ge 0\ \forall i,\ \sum_i x_i \le 1\}$.

Step 1 (volume). T_d is the cone with apex e_d and height 1 over the base $T_d \cap \{x_d = 0\}$, a copy of T_{d-1}. Its slice at height t is a copy of $(1-t)T_{d-1}$, so $|T_d| = |T_{d-1}| \int_0^1 (1-t)^{d-1}dt = |T_{d-1}|/d$. With $|T_1| = 1$ this gives $|T_d| = 1/d!$.

Step 2 (facets). A d-simplex has $d+1$ facets. For T_d they are:

- the d coordinate facets $T_d \cap \{x_i = 0\}$. Each is a copy of T_{d-1}, with area $1/(d-1)!$ by Step 1 and outward unit normal $-e_i$. Its area-normal vector is $-e_i/(d-1)!$.

- the facet $F_0 = T_d \cap \{\sum_i x_i = 1\}$, with outward unit normal $w/\sqrt{d}$. Dropping the last coordinate is the orthogonal projection onto $e_d^\perp$. It maps F_0 bijectively onto $\{x_1, \dots, x_{d-1} \geq 0,\ \sum_{i<d} x_i \leq 1\}$, a copy of T_{d-1}. By Step 1 of the proof of Lemma 2 this projection scales area by $|\langle w/\sqrt{d}, e_d\rangle| = 1/\sqrt{d}$. Hence $s_{F_0}/\sqrt{d} = 1/(d-1)!$, i.e. $s_{F_0} = \sqrt{d}/(d-1)!$, and the area-normal vector is $s_{F_0}\, w/\sqrt{d} = w/(d-1)!$.

Step 3 (shape of ΠT_d). By (4), and since $[-v/2, v/2]$ is unchanged when v is replaced by $-v$,

$$\Pi T_d = \frac{1}{(d-1)!}\Big(\sum_{i=1}^{d} [-\tfrac{e_i}{2}, \tfrac{e_i}{2}] + [-\tfrac{w}{2}, \tfrac{w}{2}]\Big).$$

Each centered segment $[-v/2, v/2]$ is the translate $-v/2 + [0, v]$, and $\sum_i [0, e_i] = [0,1]^d$. Therefore ΠT_d is a translate (by $-w/(d-1)!$) of

$$\frac{[0,1]^d + [0,w]}{(d-1)!}.$$

Step 4 (volume of the cube plus a segment). Let $Q = [0,1]^d$ and $\omega = w/\|w\|$. *Claim:* $|Q + [0,w]| = d+1$. For $y \in w^\perp$, the fiber $\{t \in \mathbb{R} : y + t\omega \in Q\}$ is empty if $y \notin \mathrm{proj}_{w^\perp} Q$, and otherwise a compact interval $[\alpha(y), \beta(y)]$. A point of $Q + [0,w]$ is $x + \sigma w$ with $x \in Q$, $\sigma \in [0,1]$; it has the same projection to $w^\perp$ as x, and its ω-coordinate is that of x plus $\sigma\|w\|$. So the fiber of $Q + [0,w]$ over y is empty for $y \notin \mathrm{proj}_{w^\perp} Q$ and equals $[\alpha(y), \beta(y) + \|w\|]$ otherwise. By Fubini,

$$|Q + [0,w]| = \int_{\mathrm{proj}_{w^\perp} Q} (\beta(y) - \alpha(y) + \|w\|)\, dy = |Q| + \|w\| \,\mathrm{vol}_{d-1}(\mathrm{proj}_{w^\perp} Q) = 1 + h_{\Pi Q}(w),$$

the last step by the definition of $h_{\Pi Q}$ and its homogeneity. The cube has $2d$ facets, each of area one, with normals $\pm e_i$, and $|\langle \pm e_i, w\rangle| = 1$. So (3) gives $h_{\Pi Q}(w) = \frac{1}{2} \cdot 2d = d$, and the claim follows. Figure 1 shows the case $d = 2$.

Step 5 (the value). By Steps 3 and 4, $|\Pi T_d| = (d+1)/((d-1)!)^d$. With Step 1 and $d! = d\,(d-1)!$,

$$R_d(T_d) = \frac{(d+1)(d!)^{d-1}}{((d-1)!)^d} = \frac{(d+1)d^{d-1}}{(d-1)!} = \frac{(d+1)d^d}{d!} = c_d.$$

Sanity check, $d = 2$: $|\Pi T_2| = 3$, $|T_2| = \frac{1}{2}$, $R_2(T_2) = 6 = c_2$.

Step 6 (all simplices). A full-dimensional simplex $\mathrm{conv}(p_0, \dots, p_d)$ equals $p_0 + LT_d$, where L is the linear map with $Le_i = p_i - p_0$; L is invertible because the p_i are affinely independent. By Lemma 4 (Part A and translations), its value is $R_d(T_d) = c_d$. □

7 The counterexample and exponential excess

7.1 Proof of the main theorem

Proof of Theorem 1. **Step 1.** T_{10} is a full-dimensional convex polytope in $\mathbb{R}^{10}$, so $K = T_{10} \times T_{10}$ is a full-dimensional convex polytope in $\mathbb{R}^{20}$, in particular a convex body.

Step 2. Proposition 3 with $r = s = 10$, $A = B = T_{10}$, and then Proposition 5 with $d = 10$, give

$$R_{20}(K) = R_{10}(T_{10})^2 = c_{10}^2 = \left(\frac{11 \cdot 10^{10}}{10!}\right)^2.$$

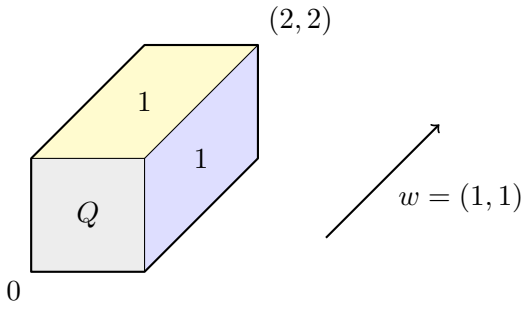


Figure 1: The planar extrusion $Q + [0, (1, 1)]$, with $Q = [0, 1]^2$, consists of the unit square and two parallelograms of area one. Its area is 3. The fiber argument in Proposition 5 proves the corresponding volume $d + 1$ in every dimension.

Step 3. Since $c_{20} = 21 \cdot 20^{20}/20!$, $10^{20}/20^{20} = 2^{-20}$ and $20!/(10!)^2 = \binom{20}{10}$,

$$\frac{R_{20}(K)}{c_{20}} = \frac{121 \cdot 10^{20} \cdot 20!}{(10!)^2 \cdot 21 \cdot 20^{20}} = \frac{121\binom{20}{10}}{21 \cdot 2^{20}}.$$

Step 4. $\binom{20}{10} = 184\,756$ and $2^{20} = 1\,048\,576$, so the numerator is $121 \cdot 184\,756 = 22\,355\,476$ and the denominator is $21 \cdot 1\,048\,576 = 22\,020\,096$. The exact integer certificate is

$$121 \cdot 184\,756 - 21 \cdot 1\,048\,576 = 335\,380 > 0.$$

Hence $|\Pi K| > c_{20}|K|^{19}$, a strict violation of (2). By Proposition 5 with $d = 20$, c_{20} is the value of every 20-dimensional simplex. □

7.2 The general product test

For $r, s \geq 2$, Propositions 3 and 5 give $R_{r+s}(T_r \times T_s) = c_r c_s$. Dividing by $c_{r+s} = (r + s + 1)(r + s)^{r+s}/(r + s)!$ and using $(r + s)!/(r!\, s!) = \binom{r+s}{r}$,

$$\frac{R_{r+s}(T_r \times T_s)}{c_{r+s}} = \frac{(r+1)(s+1)}{r+s+1}\binom{r+s}{r}\frac{r^r s^s}{(r+s)^{r+s}}.$$

Thus any universal upper constant in dimension $r + s$ must be at least $c_r c_s$. *Sanity checks:* for $r = s = 2$ the ratio is $\frac{9}{5} \cdot 6 \cdot \frac{1}{16} = \frac{27}{40} < 1$, and for $r = s = 9$ it is $\frac{100}{19} \cdot \frac{48\,620}{262\,144} < 1$, so these products do not violate (2). For $r = s = 10$ the formula reproduces Theorem 1.

7.3 Exponential excess in large dimensions

Corollary 6. *There exist a constant $\lambda > 1$ and an integer $N \geq 20$ such that, for every integer $n \geq N$, some full-dimensional convex polytope $K_n \subset \mathbb{R}^n$ satisfies*

$$\frac{R_n(K_n)}{R_n(T_n)} \geq \lambda^n.$$

The polytopes K_n may be chosen as Cartesian products of simplices.

Proof. Plan. Take k factors T_{10} and one further simplex of dimension between 10 and 19 to reach dimension n. The value is c_{10}^k times a bounded factor. Bound c_n above by $(n+1)\beta^n$ with $\beta = 11/4$, and check that $c_{10} > \beta^{10}$. Then each ten-dimensional factor gains a fixed factor larger than one.

Step 1 (the bodies). Every integer $n \geq 20$ is uniquely $n = 10k + q$ with integers $k \geq 1$ and $q \in \{10, \ldots, 19\}$ (divide $n - 10$ by 10). Put

$$K_n = T_{10}^k \times T_q,$$

where T_{10}^k is the k-fold Cartesian product, in the orthogonal coordinate product space.

Step 2 (its value). *Claim:* $R_n(K_n) = c_{10}^k c_q$. By induction on k, T_{10}^k is a full-dimensional convex polytope with $R_{10k}(T_{10}^k) = c_{10}^k$. The case $k = 1$ is Proposition 5, and the step from k to $k+1$ is Proposition 3 applied to $T_{10}^k \times T_{10}$. One more application of Proposition 3, to $T_{10}^k \times T_q$, together with Proposition 5 for T_q, proves the claim. All factor dimensions are at least 2, as required.

Step 3 ($e < \beta := 11/4$). In $e = \sum_{j \geq 0} 1/j!$ the terms with $j \leq 2$ sum to $\frac{5}{2}$. For $j \geq 3$ the ratio of consecutive terms is $1/(j+1) \leq \frac{1}{4} < \frac{1}{3}$, so

$$e = \frac{5}{2} + \sum_{j=3}^{\infty} \frac{1}{j!} < \frac{5}{2} + \frac{1}{6} \sum_{j=0}^{\infty} 3^{-j} = \frac{5}{2} + \frac{1}{6} \cdot \frac{3}{2} = \beta.$$

Step 4 (upper bound for c_n). The term $n^n/n!$ of the series $e^n = \sum_j n^j/j!$ is strictly smaller than the full sum of positive terms. By Step 3,

$$c_n = (n+1)\frac{n^n}{n!} < (n+1)e^n < (n+1)\beta^n \qquad (n \geq 2).$$

Step 5 (the ten-dimensional block). Since $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$ and $10^{10} = 2^{10}5^{10}$,

$$c_{10} = \frac{11 \cdot 10^{10}}{10!} = \frac{11 \cdot 2^2 \cdot 5^8}{3^4 \cdot 7} = \frac{17\,187\,500}{567}.$$

With $\beta^{-10} = 4^{10}/11^{10}$,

$$\Theta := \frac{c_{10}}{\beta^{10}} = \frac{2^{22} \cdot 5^8}{3^4 \cdot 7 \cdot 11^9} = \frac{1\,638\,400\,000\,000}{1\,336\,956\,340\,797} > 1,$$

using $2^{22}5^8 = 2^{14} \cdot 10^8$ and $567 \cdot 11^9 = 567 \cdot 2\,357\,947\,691$.

Step 6 (comparison). Fix λ with $1 < \lambda < \Theta^{1/10}$; this is possible since $\Theta > 1$. By Proposition 5, $R_n(T_n) = c_n$. Using Step 2, then Step 4, then $n = 10k + q$,

$$\frac{R_n(K_n)}{R_n(T_n)\lambda^n} = \frac{c_{10}^k c_q}{c_n \lambda^n} > \frac{c_{10}^k c_q}{(n+1)\beta^{10k+q}\lambda^{10k+q}} = c_q \beta^{-q} \lambda^{-q} \frac{(\Theta/\lambda^{10})^k}{10k+q+1}.$$

Step 7 (choice of N). For each fixed $q \in \{10, \ldots, 19\}$, the right side tends to infinity as $k \to \infty$, because $\Theta/\lambda^{10} > 1$ and an exponential outgrows a linear function. So there is $k_q \geq 1$ such that the right side is at least 1 for all $k \geq k_q$. Let $N = \max_q(10k_q + q)$, which is at least 20. If $n \geq N$ and $n = 10k + q$, then $10k + q \geq 10k_q + q$, so $k \geq k_q$ and $R_n(K_n)/R_n(T_n) > \lambda^n$. □

8 Verification checklist

A sceptical reader should check these load-bearing points first.

1. **The factor $\frac{1}{2}$ in the facet formula.** Almost every point of the shadow is covered by exactly two projected facets (Lemma 2, Step 2), and a projected facet has volume $s_F|\langle \nu_F, u\rangle|$ (Step 1).

2. **Facets of a product and the exponent count.** The facets of $A \times B$ are $F \times B$ and $A \times G$, with areas $s_F|B|$ and $|A|s_G$ (Proposition 3, Steps 1–2), and the powers of $|A|, |B|$ cancel as claimed (Step 4).

3. **The slanted facet of the simplex.** Its area-normal vector is $w/(d-1)!$ (Proposition 5, Step 2), which makes ΠT_d a translate of $([0,1]^d + [0,w])/(d-1)!$ (Step 3).

4. **The volume $|[0,1]^d + [0,w]| = d+1$.** Fiber argument and $h_{\Pi Q}(w) = d$ (Proposition 5, Step 4); compare Figure 1 for $d = 2$.

5. **The integer inequality.** $121 \cdot 184\,756 = 22\,355\,476 > 22\,020\,096 = 21 \cdot 2^{20}$ (proof of Theorem 1, Steps 3–4).

6. **For "all simplices have value c_d".** The area transformation $s_{LF} = |\det L|\,\|L^{-t}\nu_F\|\,s_F$ (Lemma 4, A2). Part B of that lemma is not needed for the theorem.

7. **For the corollary.** $e < 11/4$ (Step 3), $\Theta = c_{10}/(11/4)^{10} > 1$ (Step 5), and the algebra in Step 6.

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